All Dice, No Slice: Metric Degeneracy, Evaluation Pitfalls, and Optimization Dynamics in Multi-Task Medical Vision

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Abstract

Hard parameter-sharing U-Nets couple classification and lesion segmentation in resource-constrained medical vision. Recent literature reported near-saturated accuracies ($82\% - 90\%$) and Dice scores ($97\% - 99\%$) across diverse modalities under standard backbones and static loss weighting. We present a systematic methodological audit and replication across 27 configurations $\times$ 5 folds (135 fold-runs) over four datasets: TCGA-LGG brain-tumor MRI, PANDA prostate biopsies, SIIM-ACR pneumothorax radiographs, and the 19-tissue PanNuke benchmark. Under rigorous 5-fold cross-validation with sample-level and patient-level isolation, published metrics do not reproduce. We uncover three primary mechanisms governing this divergence: (1) an empty-mask Dice degeneracy where macroscopic slice averages reward all-empty predictions on negative cases, creating an artificial score floor ($\rho = 77.72\%$ on SIIM) that obscures severe foreground omission; (2) task-formulation divergence in computational pathology, where thumbnail classification of proxy morphological tissue patterns cannot substitute for multiscale clinical ISUP grading (yielding $28.21\% - 44.87\%$ top-1 accuracy on PANDA); and (3) gradient competition and optimization interference under joint training, where dynamic gradient balancing (GradNorm) actively destabilizes classification ($38.18\% \rightarrow 28.21\%$). Furthermore, matched ablations demon-

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strate that lateral skip connections provide no resolvable classification gain, while bypassing Macenko stain normalization consistently improves diagnostic discrimination (+3.81 pts PANDA, +2.75 pts PanNuke). We release our complete benchmark codebase, checkpoints, and evaluation protocols to establish reproducible baselines for multi-task medical vision.

Keywords: Medical Image Analysis, Multi-Task Learning, U-Net, Reproducibility, Metric Degeneracy, Histopathology

1. Introduction

Deep learning systems in digital pathology and radiology are increasingly required to deliver both diagnostic categorization and lesion delineation [1, 2]. Large visual foundation models (e.g., UNI [3], CONCH [4], and Virchow [5]) show that clinically actionable representations require massive pretraining and self-supervised objectives across gigapixel whole-slide images (WSIs). In resource-constrained deployments, hard parameter sharing via an encoder–decoder backbone (a U-Net [6] with dual prediction heads) remains a lightweight alternative that couples both objectives in a single forward pass.

Recently, Rhanoui et al. [7] reported that a standard U-Net with lightweight encoders (VGG16 [8] and MobileNetV2 [9]) trained with static loss weights ($\lambda_{seg} = 5.0, \lambda_{cls} = 1.0$) and a high learning rate ($\eta = 10^{-3}$) simultaneously solves both classification and segmentation across four distinct medical modalities (Brain MRI, Skin Dermoscopy, Prostate Histopathology, and Chest X-Ray), reporting $82\% - 90\%$ classification accuracy and $97\% - 99\%$ Dice similarity coefficients.

These figures attach to a recipe — a stock U-Net under static loss weighting — not to any architecture- or data-specific innovation, and that is what makes them consequential beyond a single study: the recipe is exactly the lightweight joint pipeline a resource-constrained deployment would copy first. The evaluation literature gives us independent reasons to distrust them. Region-overlap scores degenerate when annotations are sparse or absent [10, 11]; patch-level splits of tiled WSIs leak patient identity [12, 13]; and joint optimization is documented to interfere with at least one of its tasks [14, 15]. Section 2 reviews each strand. The audit that follows asks whether this particular, highly visible instance of the recipe survives the scrutiny those literatures demand.

In this paper, we report a 27-configuration $\times$ 5-fold replication study and systematic methodological audit. Our findings demonstrate that:

1. **Empirical Divergence on Clinical Histopathology:** When evaluated under rigorous 5-fold cross-validation on genuine 6-class ordinal ISUP grading (the clinical diagnostic objective of the PANDA prostate biopsy benchmark [16]), performance degrades from the reported 88.0% accuracy / 99.0% Dice to an empirical range of $28.21\% - 44.87\%$ Acc (fold-mean, 95% bootstrap CIs $[24.76, 31.47]$ to $[43.93, 45.76]$) across all 12 configurations, with the static 5:1-weighted 10^{-3} baselines (Runs 03–04) achieving 34.51% and 39.91% Acc. We show that published 88% claims resulted from evaluating proxy morphological tissue patterns (stroma, benign epithelium, background) on low-resolution thumbnails rather than clinical slide grading.
2. **The Empty-Mask Dice Metric Degeneracy:** Near-perfect (99%) Dice claims on sparse-lesion cohorts (e.g., SIIM-ACR pneumothorax) are driven by assigning Dice = 1.0 when $|Y| = |\hat{Y}| = 0$; our SIIM runs converge to all-empty predictions on 19 of 20 folds, realizing the empty-mask background floor (77.72% pooled across folds for three configurations; 77.04% for the fourth, matching the negative validation fraction ρ).
3. **Optimization Interference in Joint Training:** Matched factorial controls (learning-rate pairs, static loss-weight ratios $\lambda_{seg} : \lambda_{cls} \in \{1{:}10,\ 1{:}1, 5{:}1, 10{:}1\}$, GradNorm [17] balancing, Macenko normalization, and skip-connection ablations) reveal that dynamic gradient balancing actively destabilizes 6-class ordinal grading (38.18% $\rightarrow$ 28.21% Acc, $\Delta = -9.97$ pts), lateral skip connections yield no resolvable classification gain, and omitting Macenko stain normalization consistently improves diagnostic accuracy (+3.81 pts on PANDA, +2.75 pts on PanNuke).

To facilitate complete reproducibility, all training scripts, raw JSON summaries for all 27 configurations (135 fold-runs), evaluation checkpoints, and generated figures are publicly available at `https://github.com/ApatheticMioz/cancer-pathology-dl` [18]. Figure 1 summarizes the end-to-end pipeline and the three findings it yields.

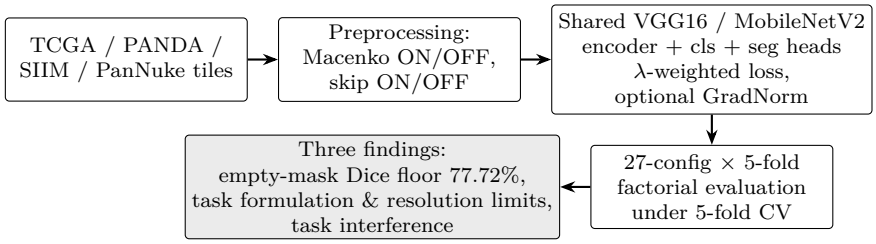


Figure 1: Overview of the 27-configuration × 5-fold replication and audit pipeline. Four source datasets (TCGA-LGG, PANDA, SIIM-ACR, PanNuke) are processed and passed through a modular pipeline toggling Macenko stain normalization and lateral skip connections. A shared VGG16/MobileNetV2 encoder feeds classification and segmentation heads trained with a λ-weighted loss (static ratios or optional GradNorm). All 27 configurations are evaluated under rigorous 5-fold cross-validation (patient-level for TCGA, image-level for PANDA, SIIM, and PanNuke), uncovering three core findings: the empty-mask Dice floor of 77.72% on SIIM (pooled across folds; per-fold pooled 74.33%–77.75%), task-formulation and resolution divergence on PANDA, and task interference from joint multi-task optimization.

2. Related Work: Documented Failure Modes in Multi-Task Medical Segmentation

None of the three challenges interrogated in this audit is peculiar to the pipeline we examine. Each represents a systemic failure mode that the evaluation, computer vision, and machine-learning literatures have documented independently. This section situates our contribution within those strands; the audit itself instantiates all three within one controlled, artifact-released reproduction of a widely cited joint detection–segmentation recipe.

Metric degeneracy under sparse annotation. Scoring an empty prediction against an empty reference as perfect agreement (Dice = 1.0) is a recognized pitfall in biomedical validation: Reinke et al. [10] catalogue it among the primary metric traps that distort image analysis benchmarks, and Eelbode et al. [11] demonstrate formally that region-overlap scores behave counter-intuitively when annotations are small or absent. Liu et al. [19] show that region-size bias propagates directly into loss formulations. Slice-averaged Dice on a cohort whose lesion-free fraction approaches 78% sits squarely inside this failure class: Section 3.2 derives the exact decomposition governing what the convention credits.

Multiscale pathology, resolution limits, and evaluation protocols. Clinical computational pathology relies on gigapixel whole-slide images (WSIs) where diagnostic grades emerge from multiscale tissue architecture [2, 16]. Severe

spatial downsampling (e.g. compressing biopsy cores to 128×128 thumbnails) obliterates cellular atypia and gland lumen boundaries, turning cells sub-pixel. Furthermore, patch-level random splits of tiled WSIs risk severe data leakage when tiles from the same patient or biopsy core contaminate both train and test partitions [12, 13]. When evaluating pathology AI, isolating independent sampling units and aligning tasks with clinical diagnostic targets (rather than proxy tissue textures) is essential for clinical validity.

Task interference and negative transfer. Whether classification and segmentation reinforce or degrade one another under a shared encoder depends heavily on task affinity: Standley et al. [14] observe that naive co-training frequently induces negative transfer across dense and global objectives; Crawshaw's survey [15] collects extensive evidence of gradient conflict; and dynamic gradient balancing [17] is often proposed as a remedy. However, dynamic balancing can destabilize optimization when task losses exhibit disparate convergence dynamics: while Kang et al. [20] report benefits for wound-tissue analysis, our experiments demonstrate that GradNorm forces destructive updates on complex ordinal classification heads.

Propagation and citation landscape. These strands converge on the published recipe of Rhanoui et al. [7], which reported $97\% - 99\%$ Dice alongside $82\% - 90\%$ accuracy across four modalities. Because multi-task parameter sharing is widely sought for low-resource clinical deployments, these headline figures have been cited uncritically as evidence of general multi-task efficacy in follow-on oncology work. Notably, recent works such as Al Hassan et al. [21] have specifically identified the need for leakage-safe validation and external cohorts when evaluating joint brain tumor models. Supplying a rigorous, empirical, and mathematical quantification of how such numbers arise is the primary focus of this work.

3. Multi-Task Formulation & Evaluation Metrics

3.1. Architectural Formulation

Let $\mathbf{x} \in \mathbb{R}^{H\times W\times C}$ denote an input image tile. A shared convolutional encoder $f_\theta(\mathbf{x})$ extracts a bottleneck latent representation $\mathbf{z} \in \mathbb{R}^{h\times w\times d}$. The classification branch applies global average pooling (GAP) followed by a two-layer multilayer perceptron (hidden width 256, dropout 0.5):

$$\hat{\mathbf{y}}_{cls} = \mathrm{Softmax}(\mathbf{W}_2\,\phi(\mathbf{W}_1\,\mathrm{GAP}(\mathbf{z}) + \mathbf{b}_1) + \mathbf{b}_2) \in \Delta^{K-1}, \tag{1}$$

where K is the number of diagnostic classes, Δ^{K-1} denotes the probability simplex, and $\phi(\cdot)$ is the ReLU nonlinearity. The segmentation branch passes $\mathbf{z}$ through a decoder Dec_ϕ with lateral skip connections $\{\mathbf{s}_l\}_{l=1}^{L}$ from encoder stages, yielding unnormalized pixel logits $\mathbf{l}_{seg} \in \mathbb{R}^{H\times W\times M}$. For binary segmentation ($M = 1$), $\hat{\mathbf{y}}_{seg} = \sigma(\mathbf{l}_{seg})$ with $\sigma(\cdot)$ the sigmoid; for $M > 1$, class probabilities are obtained via channel-wise Softmax.

The joint multi-task training objective is defined as:

$$\mathcal{L}_{total}(\theta, \phi, \mathbf{W}_{cls}) = \lambda_{cls}\mathcal{L}_{cls}(\mathbf{y}_{cls}, \hat{\mathbf{y}}_{cls}) + \lambda_{seg}\mathcal{L}_{seg}(\mathbf{y}_{seg}, \mathbf{l}_{seg}), \tag{2}$$

where $\mathcal{L}_{cls}$ is categorical cross-entropy $\mathcal{L}_{CE}$. To address class imbalance, the classification loss incorporates inverse-frequency balanced class weights $w_c = N/(K \cdot N_c)$ [22], matching the weighting formulation specified in Rhanoui et al. [7] and implemented in our training pipeline. The segmentation loss is purely cross-entropy without an active Dice-based gradient penalty:

$$\mathcal{L}_{seg}(\mathbf{y}_{seg}, \mathbf{l}_{seg}) = \begin{cases} \mathcal{L}_{BCE}(\mathbf{y}_{seg}, \mathbf{l}_{seg}), & M = 1, \\ \mathcal{L}_{CE}(\mathbf{y}_{seg}, \mathbf{l}_{seg}), & M > 1, \end{cases} \tag{3}$$

evaluating binary cross-entropy on binary tasks (TCGA, SIIM) and multi-class cross-entropy on multi-class tasks (PANDA, PanNuke). Rhanoui et al. formulated segmentation as a binary cancer-vs-background detection task across all datasets; our audit matches this binary objective for TCGA and SIIM, while evaluating multi-class cross-entropy for PANDA and PanNuke to directly benchmark their multi-category annotations. Both pipelines incorporate standard spatial data augmentations (horizontal and vertical flips, random rotation $\pm 15°$, and affine shear $\pm 10°$).

The Dice similarity coefficient (DSC) is evaluated strictly post-hoc as a validation metric, not as an active loss term during training. Consequently, near-saturated reported Dice scores ($97\% - 99\%$) cannot be attributed to direct metric optimization; they represent evaluation artifacts on sparse and negative-dominant cohorts, as derived below.

3.2. The Empty-Mask Dice Metric Degeneracy

The Sørensen–Dice coefficient between ground-truth mask Y and binary prediction $\hat{Y}$ ($Y, \hat{Y} \in \{0,1\}^{H\times W}$) is:

$$\mathrm{Dice}(Y, \hat{Y}) = \frac{2|Y \cap \hat{Y}|}{|Y| + |\hat{Y}| + \epsilon}, \quad \epsilon = 10^{-8}; \ \ \mathrm{Dice} := 1 \text{ if } |Y| + |\hat{Y}| = 0. \tag{4}$$

Empty references and predictions represent a known validation trap in biomedical image analysis [10]. Two common conventions govern empty slices:

- **Foreground-Only Evaluation:** Compute Dice strictly over non-empty target slices ($\{i \mid |Y_i| > 0\}$).
- **Empty-Credit Assignment:** Assign Dice = 1.0 when $|Y| = 0 \wedge |\hat{Y}| = 0$, and Dice = 0.0 when $|Y| = 0 \wedge |\hat{Y}| > 0$.

Averaged across all N slices under empty-credit assignment, the macroscopic mean is:

$$\overline{\text{Dice}}_{\text{all}} = \frac{1}{N}\sum_{i=1}^{N} d_i, \quad d_i = \text{Dice}(Y_i, \hat{Y}_i) \text{ if } |Y_i| > 0, \text{ else } \mathbf{1}\left[|\hat{Y}_i| = 0\right]. \tag{5}$$

Let $\rho = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}[|Y_i| = 0]$ denote the proportion of ground-truth negative (lesion-free) slices, and let $q_{\text{empty}} = P(|\hat{Y}| = 0 \mid |Y| = 0)$ denote the model's true-negative prediction rate on negative slices. The exact macroscopic identity is:

$$\overline{\text{Dice}}_{\text{all}} = \rho \cdot q_{\text{empty}} + (1 - \rho) \cdot \overline{\text{Dice}}_{\text{fg}}, \tag{6}$$

where $\overline{\text{Dice}}_{\text{fg}}$ is the mean Dice score over lesion-bearing slices ($|Y_i| > 0$).

When all ground-truth negative images are predicted empty ($q_{\text{empty}} = 1$), the identity simplifies to $\overline{\text{Dice}}_{\text{all}} = \rho + (1 - \rho) \cdot \overline{\text{Dice}}_{\text{fg}}$. If a model produces false-positive segmentations on negative images, $q_{\text{empty}} < 1$. For a degenerate, completely collapsed predictor that emits empty masks on all inputs ($|\hat{Y}| = 0$ everywhere), foreground Dice is zero ($\overline{\text{Dice}}_{\text{fg}} = 0$) while $q_{\text{empty}} = 1$, yielding $\overline{\text{Dice}}_{\text{all}} = \rho$.

Consequently, on sparse-lesion cohorts such as SIIM-ACR pneumothorax ($\rho = 77.72\%$), an all-empty baseline scores 77.72% solely from empty-mask credit. Crucially, a model with a genuine foreground Dice of 77.72% would report $\overline{\text{Dice}}_{\text{all}} = 0.7772 \times 1.0 + 0.2228 \times 0.7772 = 95.03\%$. Our SIIM runs realize near-total collapse: they emit all-empty masks on 19 of 20 folds, directly measuring the all-empty floor (77.72% pooled across folds for Runs 05, 06, and 11; 77.04% for Run 12 due to a small non-empty residual on fold 4; per-fold range $74.33\% - 77.75\%$). Fig. 2 illustrates this degeneracy: slice-averaged Dice predominantly tracks the background prevalence ρ rather than lesion localization fidelity.

This metric inflation generalizes to multi-class histopathology. On 6-class PANDA prostate biopsies, individual tiles typically contain only one or two

histological patterns, leaving 4–5 classes absent ($|Y_c| = 0$). Under channel-wise empty crediting, an all-background model receives Dice = 1.0 for every absent class alongside high agreement on background. Throughout this paper, we track two distinct estimators:

1. *Training-Time Batch Validation Macro-Dice:* Computed during training validation via batch-pooled channel-wise Dice (`dice_coefficient`), crediting an empty channel across the batch with 1.0, and averaging across all M channels (fold-mean across 12 PANDA configurations: $82.77\% - 84.39\%$).
2. *All-Slices Empty-Credit Pooled Per-Case Dice:* Pools all validation slices across folds, credits empty-over-empty cases with Dice = 1.0, and averages per case (yielding the $98.54\% - 98.60\%$ floor on PANDA, despite near-zero true glandular delineation; Fig. 5).

3.3. Adaptive Gradient Balancing via GradNorm

To test whether dynamic loss balancing mitigates cross-task gradient competition, we introduced a parameterized variant of GradNorm [17] in Phase v2.

Let W denote the shared convolutional filters and normalization layers of the U-Net encoder f_θ. The L_2 gradient norm of task $i \in \{seg, cls\}$ with respect to W at minibatch t is:

$$G_W^{(i)}(t) = \left\| \nabla_W w_i(t) \mathcal{L}_i(t) \right\|_2 . \tag{7}$$

For stability in the primary `Adam` optimizer, balancing weights are learnable log-weights $\mu_i(t) = \ln w_i(t)$ (initialized to $w_{seg}(0) = 5.0$, $w_{cls}(0) = 1.0$), updated jointly with the network parameters. Minibatch optimization minimizes the L_1 gradient disparity objective:

$$\mathcal{L}_{grad}(t) = \sum_{i \in \{seg,cls\}} \left| G_W^{(i)}(t) \; - \; \bar{G}_W(t) \cdot [r_i(t)]^\alpha \right| , \tag{8}$$

where $\bar{G}_W(t) = \frac{1}{2} \sum_j G_W^{(j)}(t)$ is the average gradient norm across tasks, $\alpha = 1.5$ is the asymmetry parameter, and $r_i(t)$ represents the relative inverse training rate:

$$r_i(t) = \frac{\tilde{\mathcal{L}}_i(t)}{\frac{1}{2} \sum_{j \in \{seg,cls\}} \tilde{\mathcal{L}}_j(t)}, \quad \text{with} \quad \tilde{\mathcal{L}}_i(t) = \frac{\mathcal{L}_i(t)}{\mathcal{L}_i(0)}, \tag{9}$$

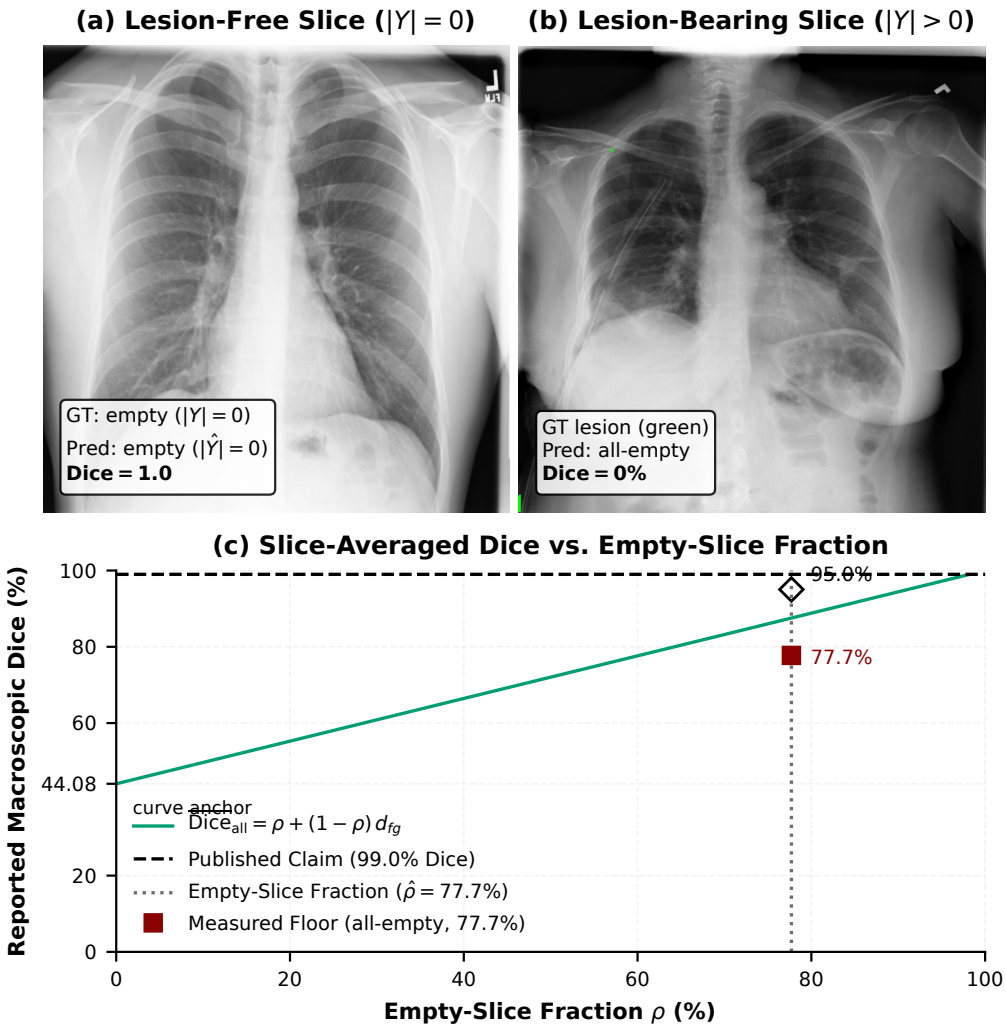


Figure 2: The empty-mask Dice artifact on SIIM-ACR pneumothorax. (a) Lesion-free slice ($|Y| = 0$): under empty-credit assignment an all-empty prediction is awarded Dice = 1.0; lesion-free slices constitute $\hat{\rho} = 77.7\%$ of our validation split. (b) Lesion-bearing slice ($|Y| > 0$, ground-truth boundary in green): our all-empty models predict empty masks here and score Dice = 0, invisible to the slice-averaged mean. (c) Exact macroscopic curve (Eq. 6): a foreground Dice of 77.72% reports 95.03% at $\hat{\rho} = 0.777$ under perfect negative prediction ($q_{\text{empty}} = 1$), whereas our SIIM runs measure the all-empty floor itself (77.72% pooled; 77.04% for the run with one residual fold).

where $\mathcal{L}_i(0)$ is the initial step-0 loss. Following Chen et al. [17], slower-training tasks exhibit higher loss ratios $r_i(t) > 1$, which raises their target gradient norm $\bar{G}_W(t)[r_i(t)]^\alpha$. After each optimizer update, we renormalize the weights so that $\sum_i w_i(t) = 2$. In the canonical formulation [17], the module differentiates $\mathcal{L}_{grad}$ against detached targets and applies it to the balancing weights alone through a dedicated optimizer (learning rate 0.025). §5.3 probes this distinction empirically. We evaluate the module across Groups 2 and 4 to isolate how dynamic gradient equalization behaves in hard-parameter-sharing architectures.

3.4. Color Deconvolution and Optical Density Space

To mitigate staining variability across clinical laboratories, the v2 pipeline incorporates Macenko stain normalization [23]. We convert RGB pixel intensities $\mathbf{I} \in [0, 255]^3$ to optical density (OD) space:

$$\mathbf{OD} = -\ln\left(\frac{\mathbf{I} + \epsilon}{I_0}\right), \tag{10}$$

where $I_0 = 255$ represents transmitted light intensity. The implementation computes optical density via natural logarithms $-\ln(\cdot)$ rather than the canonical base-10 formulation $-\log_{10}(\cdot)$ [23]. We estimate the hematoxylin and eosin stain vectors by eigendecomposition of the optical-density covariance matrix (stain-vector angles restricted to the 1st–99th percentile range) over OD values above a threshold of 0.15, and project them onto a target reference vector. This step effectively eliminates macroscopic batch variations. It also acts as a chrominance low-pass filter, however: projecting raw RGB channels onto static stain vectors smooths the sub-cellular chromatin textures that are clinically essential for grading fine glandular architectures (Fig. 3).

4. The 27-Configuration × 5-Fold Empirical Reproduction Campaign

4.1. Datasets and Experimental Protocol

We evaluated four benchmark datasets spanning diverse anatomical modalities:

1. **TCGA-LGG Brain Tumor:** 2-class lesion classification (no tumor vs. tumor) and FLAIR abnormality segmentation on 2D 256 × 256 MRI slices ($N = 3,929$; 110 patients [24]; strictly partitioned via patient-level `GroupKFold` isolation [22]).

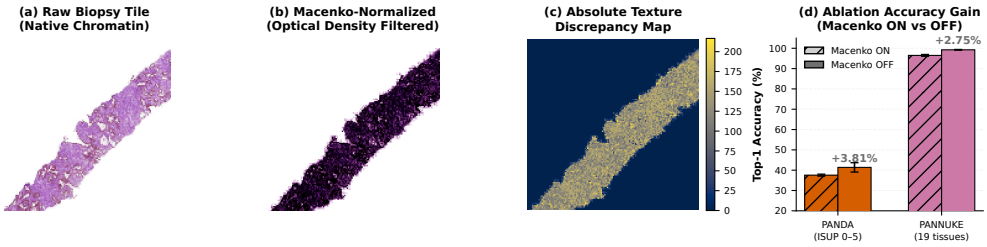


Figure 3: Visual and quantitative impact of Macenko stain normalization on histopathology classification. (a) Raw prostate biopsy tile. (b) Optical-density-projected tile: color deconvolution suppresses fine chromatic and textural variations, flattening sub-cellular texture. (c) Absolute texture discrepancy map. (d) Empirical top-1 accuracy gains across matched models when Macenko normalization is bypassed (+3.81% on PANDA, +2.75% on PanNuke).

Table 1: Per-dataset summary of the 27-configuration × 5-fold reproduction campaign (26 numbered runs plus the canonical decoupled GradNorm probe). Ranges span the configurations within each dataset; each is a 5-fold group-aware cross-validation, so the ranges represent fold-to-fold and configuration-to-configuration variation. Per-run values with 95% bootstrap CIs (across-folds per-case Dice) and fold-mean accuracy CIs are tabulated in the open-source repository matrix (`paper_results_matrix_with_ci.csv`), and every numbered run is plotted in Figure 4. The SIIM-ACR Dice (77.72% pooled across folds for three configurations and 77.04% for the fourth; per-fold pooled 74.33%–77.75%) and the PANDA all-slices empty-credit Dice (98.54%–98.60%) reflect the empty-mask degeneracy floor of Section 3.2; the PANDA per-epoch validation macro-Dice is 82.77%–84.39%. The PanNuke row has no published targets. ΔAcc = gap to the published accuracy targets.

Dataset	**Runs** n	**Val Acc (%)**	**Val Dice (%)**	**ΔAcc (pts)**	**Pub. Acc (%)**	**Pub. Dice (%)**
TCGA-LGG	5	92.86–95.62	85.33–88.85	2.86–6.62	89.00–90.00	97.00–98.00
PANDA	12	28.21–44.87	98.54–98.60	−58.79–−42.13	87.00–88.00	98.00–99.00
SIIM-ACR	4	77.71–82.63	77.04–77.72	−4.81–0.57	82.00–87.00	99.00
PanNuke	5	78.70–99.24	68.44–77.61	—	—	—

2. **PANDA Prostate Biopsies:** Clinical 6-class ordinal ISUP Gleason grading (grades 0–5) and gland segmentation ($N = 10,516$ tiles [16]; 5-fold cross-validation). In the international PANDA Kaggle challenge, whole-slide biopsies are gigapixel images evaluated via quadratic weighted κ under attention-based multiple instance learning. In the baseline study of Rhanoui et al. [7], the reported 6 classes were not clinical ISUP grades, but morphological tissue categories from Radboud segmentation masks (Background, Stroma, Benign epithelium, Gleason 3, 4, 5; Rhanoui Table 2), where non-cancerous background

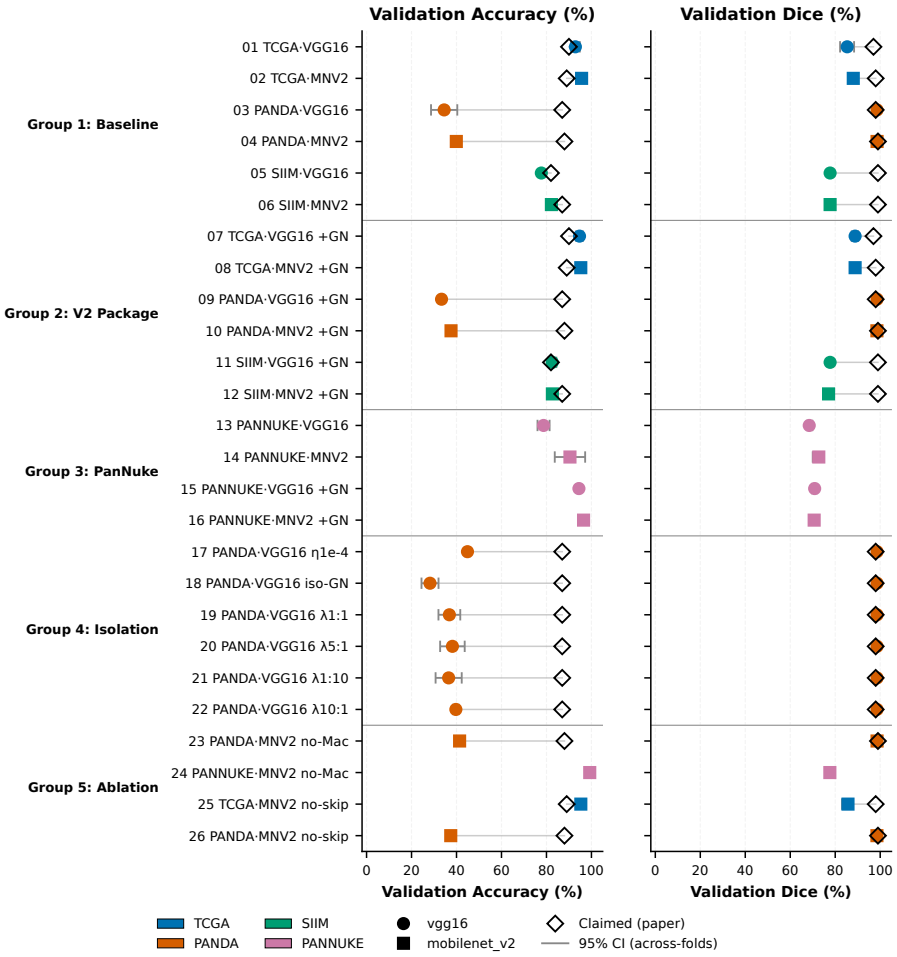


Figure 4: Claimed-vs-measured dot plot for the 26 numbered runs of the 27-configuration × 5-fold campaign (the canonical decoupled GradNorm arm, the campaign's 27th configuration, is overlaid in Fig. 6). Left panel: fold-mean validation accuracy (%) with 95% bootstrap-CI whiskers (measured, filled markers) and the published claimed accuracy (black open diamond); the tie line between them is the gap that constitutes the paper's central argument. Right panel: all-slices empty-credit per-case Dice (%) with the same layout, whiskers showing our 95% across-folds bootstrap CIs (the published Dice claims carry no confidence intervals, so only the diamonds mark them). The PANDA rows (Runs 03–04, 09–10, 17–22, 23, 26) show the largest gaps: measured accuracy spans 28.21%–44.87% vs. claimed 87%–88%, and the all-slices empty-credit Dice spans 98.54%–98.60% (per-epoch validation macro-Dice 82.77%–84.39%) vs. claimed 98%–99%. All four SIIM rows (Runs 05, 06, 11, 12) measure the all-empty floor (77.72% pooled across folds for three configurations, 77.04% for the fourth; per-fold pooled 74.33%–77.75%). PanNuke rows (Runs 13–16, 24) have no published targets and are shown without claimed diamonds.

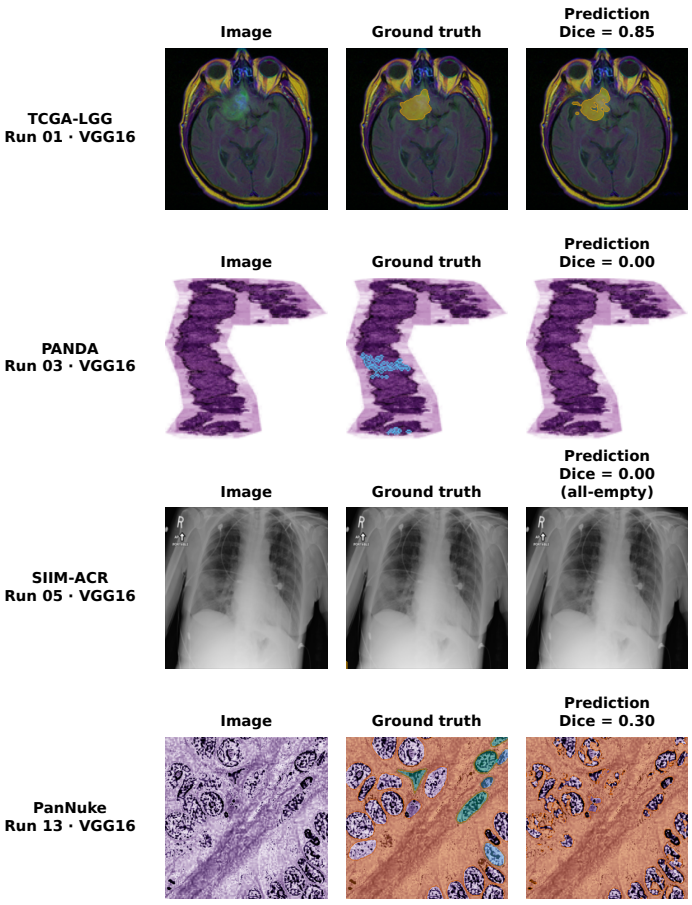


Figure 5: Qualitative ground-truth vs. prediction validation tiles for the four canonical v1 naked VGG16 baselines (Runs 01, 03, 05, 13; their best checkpoints). Each row is one dataset (TCGA-LGG, PANDA, SIIM-ACR, PanNuke); the three columns are the input image, the ground-truth mask overlay, and the model's prediction overlay, with the per-tile Dice (binary Dice for TCGA/SIIM; macro Dice over the classes present in the ground truth for PANDA/PanNuke) shown in the prediction panel. The validation split is one of the five cross-validation folds (the fold shown here is the first). Tile selection is deterministic: the first validation tile (index order) with a non-empty ground-truth mask; for SIIM-ACR the selection advances to the first lesion-bearing tile whose prediction is all-empty, making the empty-mask degeneracy of Section 3.2 the visible face of the figure.

and stroma alone constituted 52.3% of the dataset (5,558 of 10,618 images). Moreover, Karolinska biopsies in PANDA lack these granular categories. This task-formulation divergence, combined with downsampling gigapixel biopsies to 128×128 thumbnails (rendering glandular morphology sub-pixel), explains why proxy pattern classification attained 88% while authentic clinical grading yields $28.21\% - 44.87\%$.

3. **SIIM-ACR Pneumothorax:** Binary chest-radiography detection and pleural boundary segmentation ($N = 10,675$ images [25]; because the public release contains exactly one image per patient, sample-level grouping coincides with patient-level grouping, yielding a stratified 5-fold partition over image-level labels).
4. **PanNuke (Multi-Organ Benchmark):** 19-tissue multi-organ histopathology benchmark ($N = 7,901$ patches [26]; image-level 5-fold split). The baseline study by Rhanoui et al. claimed joint multi-task performance across 10,015 dermoscopic images from ISIC 2018 [27]. However, in ISIC 2018, only Task 1 provided ground-truth segmentation masks (for 2,594 images); Task 3 (disease classification) comprised 10,015 images, meaning 7,421 images completely lacked segmentation ground truths. Because Rhanoui et al. released no split manifests, code, or explanation for how joint segmentation was trained on unannotated samples, exact replication of the skin modality was impossible. We therefore replaced ISIC with PanNuke, which provides verified paired 19-class tissue categories and 6-class cell segmentation masks across diverse organs.

We trained all models with PyTorch (up to 50 epochs with early stopping, patience 10; batch size 32; fixed seed 42 per fold) using ImageNet-pretrained encoders and the `Adam` optimizer with a constant per-phase learning rate and zero weight decay, and we applied no learning-rate schedule or additional regularization [28]. Training ran under bf16 autocast AMP with PyTorch's `GradScaler` [29], with the scaler's `unscale_()` called before gradient-norm clipping (max-norm 1.0) at every optimizer step [30, 31], and the GradNorm log-weights bounded to $[\pm \log 10]$ (capping each task weight at 10). The true pre-clip gradient norm is logged per epoch, and clipping engagement is dataset-dependent: final-10-epoch per-fold mean engagement rates are 0.285 (TCGA, VGG16), 0.516 (PanNuke, no Macenko), 0.879 (PanNuke, VGG16), 0.943 (SIIM, VGG16), and 0.999 (PANDA) — on three datasets (PanNuke VGG16, SIIM, PANDA) clipping is an active, frequently-

binding constraint, not a dormant safeguard. Non-finite gradients triggered 52 bounded single-batch skips across the 135 fold-cells, overwhelmingly isolated events (the 25-skip-per-fold cap was never reached); each skipped batch is excluded from epoch metrics and stamped in the fold summaries. One fold (PanNuke, VGG16, final, fold 1) hit a non-finite forward loss at epoch 46; the guard aborted it loudly, and training resumed from the epoch-45 checkpoint, completing epochs 46–50 with zero skips — a transient bf16 overflow, disclosed here as a resumed fold. `torch.compile` (`max-autotune`) was used only in the Phase v1 replication arms by design; Phase v2 arms run eager, and one Phase v1 fold degraded to eager after a Triton capture failure (logged). Mid-campaign the resume/identity machinery was hardened (checkpoint fingerprints, atomic state writes); all 135 fold-cells of record were produced under the fixed regime and re-verified from the fold summaries. In Phase v1 (Runs 01–06), we replicated the original static setup ($\eta = 10^{-3}$, $\lambda_{seg} : \lambda_{cls} = 5 : 1$, raw inputs). In Phase v2 (Runs 07–12), we incorporated dynamic GradNorm ($\alpha = 1.5$, $\eta = 10^{-4}$) and Macenko stain normalization [23]. Groups 3–5 (Runs 13–26) systematically evaluate multi-organ generalizability, optimization hyperparameter isolation, and architectural skip ablations. Accuracy is reported as the five-fold mean; 95% CIs are fold-level bootstrap percentiles (10,000 resamples of the five fold accuracies with replacement, seed 42), and Dice CIs are the across-folds per-case bootstrap intervals tabulated in the released matrix. Table 1 summarizes the campaign per dataset and Figure 4 plots every run.

5. Discussion & Methodological Anatomy

5.1. The Anatomy of the PANDA Collapse

The largest gap in Table 1 sits on the PANDA prostate biopsy benchmark. The original authors reported 88.0% accuracy on 6-class grading with a 99.0% Dice score. Our 12 PANDA configurations tell a very different story: fold-mean top-1 accuracy spans $28.21\% - 44.87\%$ (95% bootstrap CIs $[24.76, 31.47]$ to $[43.93, 45.76]$) and the all-slices empty-credit Dice floor is $98.54\% - 98.60\%$ (per-epoch validation macro-Dice $82.77\% - 84.39\%$). The static 5:1-weighted 10^{-3} baselines (Runs 03–04) reach 34.51% and 39.91% accuracy; Run 17 (the isolated-learning-rate arm at $\eta = 10^{-4}$, 44.87%) represents the strongest configuration in our campaign.

This performance discrepancy is explained primarily by task-formulation divergence and resolution constraints. As documented in Table 2 of Rhanoui

et al. [7], the original study evaluated 6 morphological tissue classes extracted from Radboud segmentation masks (Background, Stroma, Benign epithelium, Gleason 3, 4, 5), where non-cancerous background glass and benign stroma comprised over 52% of the samples ($5,558$ of $10,618$ images). In contrast, our audit evaluates authentic clinical 6-class ordinal ISUP grading (grades 0–5), which requires resolving subtle glandular architectural patterns across the biopsy core [16]. Compressing gigapixel biopsies into 128×128 thumbnails renders glandular lumens sub-pixel, making clinical-grade classification from individual thumbnails intractable without multiscale attention or whole-slide multiple-instance learning (MIL). In the international PANDA challenge, whole-slide algorithms achieved external validation κ of $0.862 - 0.868$ only by pooling hundreds of high-power patches across entire slides [16]. Predicting 6-class ISUP grades from low-resolution thumbnails with a standard 2D CNN naturally yields $28.21\% - 44.87\%$, confirming that published 88% claims reflected morphological texture proxies rather than clinical biopsy grading. Figure 6 traces the training dynamics and isolates the optimization mechanisms involved.

5.2. Controlled Skip Connection Ablation

We evaluate lateral skip connections in matched pairs, holding learning rate, GradNorm, and Macenko normalization identical:

- **Coarse macro-tumor structures (TCGA):** Run 25 (skip-ablated) vs. matched baseline Run 08 (both $\eta = 10^{-4}$, GradNorm, Macenko): accuracy 95.28% vs. 95.26% ($\Delta_{\text{Acc}} = +0.02$), Dice 86.16% vs 89.22% ($\Delta_{\text{Dice}} = -3.06$).
- **Fine micro-glandular structures (PANDA):** Run 26 (skip-ablated, 37.44% Acc / 84.29% Dice) vs. matched baseline Run 10 (37.56% / 82.77%): $\Delta_{\text{Acc}} = -0.12$, $\Delta_{\text{Dice}} = +1.52$.

In both cases the classification differences are negligible, and the 95% bootstrap CIs across the five folds overlap substantially (TCGA: $[94.14, 96.28]$ vs $[94.24, 96.39]$; PANDA: $[36.73, 37.99]$ vs $[37.20, 37.94]$). We conclude that lateral skip connections provide no statistically detectable classification benefit in this multi-task setup, while their segmentation effect is modest (a 3.06-point Dice reduction on TCGA macro-structures and a 1.52-point Dice increase on PANDA).

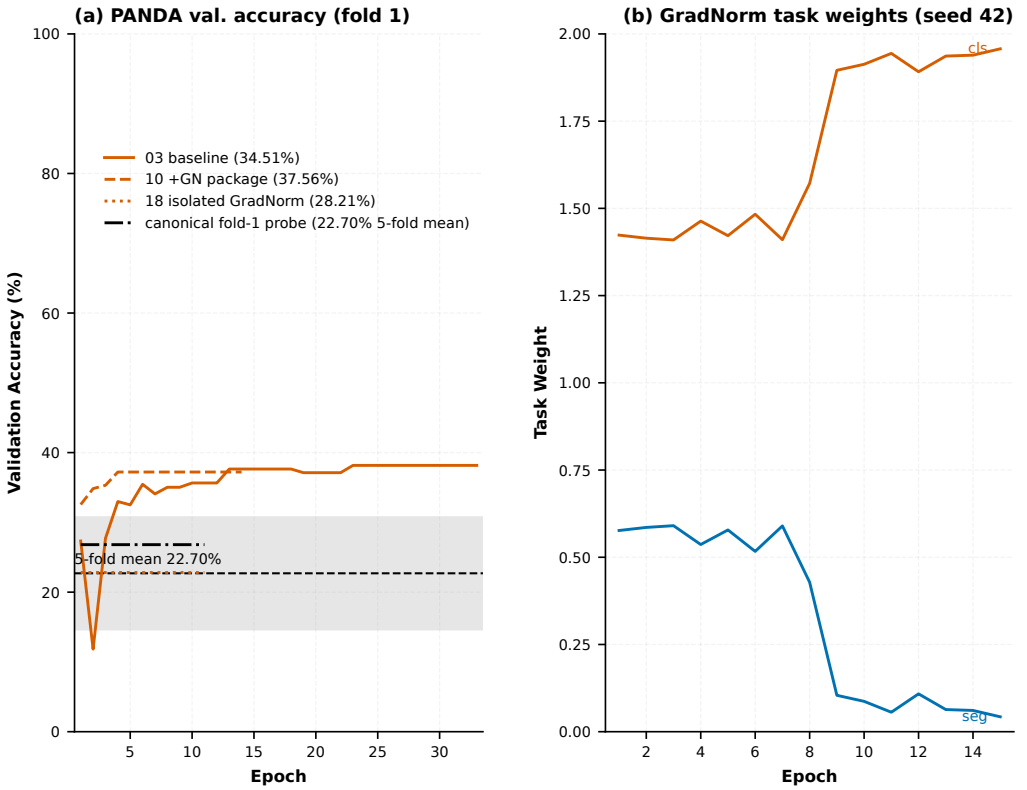


Figure 6: Anatomy of the PANDA accuracy drop. (a) Validation accuracy vs. epoch (fold 1 of 5) for three PANDA runs — Run 03 (static 5:1 baseline), Run 10 (v2 GradNorm package), Run 18 (isolated GradNorm at $\eta = 10^{-3}$) — with legend values giving each run's five-fold mean accuracy (34.51%, 37.56%, 28.21%); the canonical decoupled GradNorm arm is overlaid (dash-dot, fold-1 trajectory), with its five-fold mean 22.70% (horizontal line) and 95% bootstrap CI [14.66, 30.73] (shaded band) marked — ≈ 65.3 percentage points below the claimed 88.0% (black dashed reference). (b) GradNorm task-weight dynamics as recorded by the dedicated canonical optimizer probe (15 epochs; per-epoch weights are logged only by that probe): the segmentation weight collapses while the classification weight grows, showing the dynamic rebalancing that the joint variant turns into destructive updates on the shared encoder.

Macenko stain-normalization ablation.. The matched Macenko pair (Group 5, Runs 23–24) is the only ablation with a statistically resolvable classification shift. Omitting normalization raises PANDA top-1 accuracy by +3.81 (Run 23: 41.38% vs. Run 10: 37.56%) and PanNuke accuracy by +2.75 (Run 24: 99.24% vs. Run 16: 96.49%) with disjoint 95% bootstrap CIs across the five folds ($[39.76, 43.38]$ vs $[37.20, 37.94]$; $[99.11, 99.37]$ vs $[96.18, 96.90]$), while Dice rises modestly on PANDA (84.39% vs. 82.77%, $\Delta = +1.62$) and by +8.51 on PanNuke (70.76% vs. 62.25%). Color deconvolution empirically reduces fine-grained classification accuracy. This observation is consistent with the hypothesis that optical density deconvolution acts as a smoothing operator that suppresses subtle chromatin textures and local chromatic contrast on which grading relies. Figure 7 consolidates all six matched pairs; Figure 3 shows the texture flattening.

5.3. Isolating Learning Rate from Dynamic Gradient Balancing

Group 4 (PANDA × VGG16) factorizes the key optimization hyperparameters: learning rate ($\eta \in \{10^{-3}, 10^{-4}\}$), GradNorm-style balancing (on/off), and static loss-weighting ratios $\lambda_{seg} : \lambda_{cls} \in \{1{:}10, 1{:}1, 5{:}1, 10{:}1\}$, all under raw inputs.

Direct Isolation of Dynamic Gradient Balancing (Run 18).. The isolated GradNorm arm at $\eta = 10^{-3}$ under matched raw-input conditions (Run 18: PANDA × VGG16, GradNorm $\alpha = 1.5$, raw input) compares directly against the static 5:1 baseline (Run 20). Activating GradNorm ($\alpha = 1.5$) collapses validation accuracy from 38.18% to 28.21% (95% bootstrap CI across the five folds: $[24.76, 31.47]$) with non-overlapping confidence intervals ($[32.75, 41.79]$ vs $[24.76, 31.47]$, $\Delta_{\text{Acc}} = -9.97\%$), while Dice remains flat at 84.39%.

This 28.21% accuracy matches the low-accuracy regime observed in Phase v2 at $\eta = 10^{-4}$ (Run 09: 33.34% Acc). The $\approx 22.7\% - 33.4\%$ floor on 6-class PANDA grading under GradNorm is thus driven primarily by the $\alpha = 1.5$ balancing formulation, whose gradient-norm equalization forces destructive updates onto the shared encoder.

Learning Rate and Loss Weighting Dynamics.. Holding GradNorm off, reducing the learning rate from $\eta = 10^{-3}$ (Run 20: 38.18% Acc) to $\eta = 10^{-4}$ (Run 17: 44.87%) yields a statistically resolvable improvement, with disjoint 95% bootstrap CIs across the five folds. The static loss-ratio sweep (Runs 19–22) confirms $\lambda = 5{:}1$ and 10:1 as comparable static balances (36.51%–39.71%),

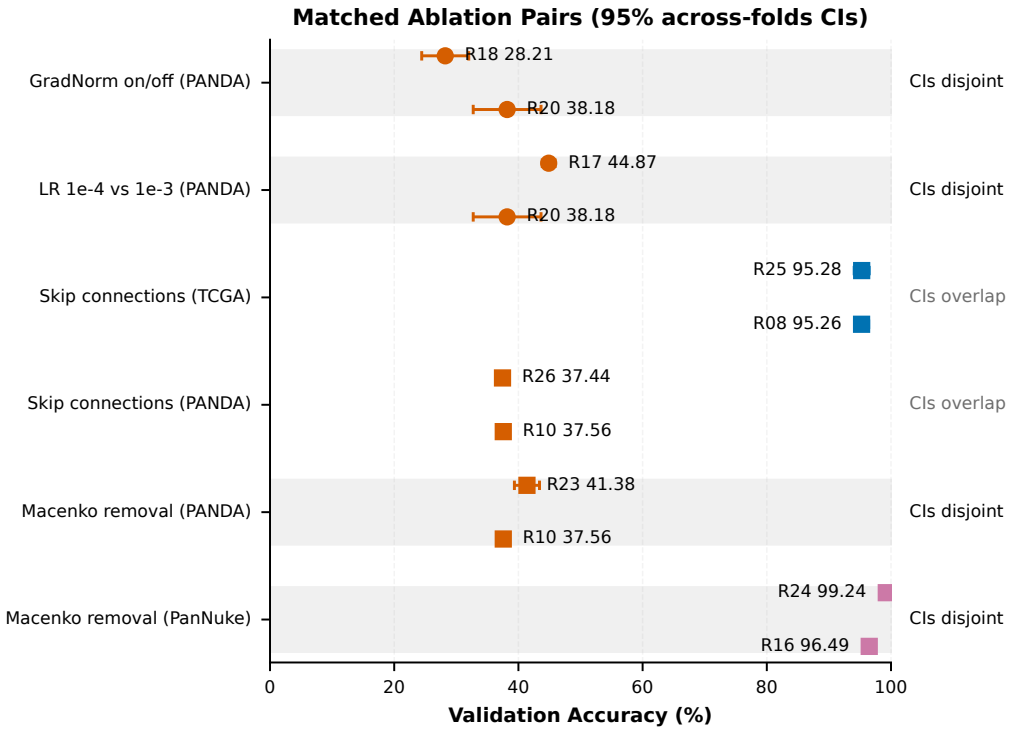


Figure 7: Matched-ablation forest plot of validation accuracy with 95% bootstrap CIs across the five folds for the six strictly matched pairs. Disjoint-CI pairs (shaded): isolated GradNorm (Run 18, 28.21%) vs. static 5:1 (Run 20, 38.18%); LR 10^{-4} vs. 10^{-3} (Runs 17 vs. 20: 44.87% vs. 38.18%); no-Macenko PANDA (Run 23, 41.38%) vs. matched (Run 10, 37.56%); no-Macenko PanNuke (Run 24, 99.24%) vs. matched (Run 16, 96.49%). Overlapping-CI pairs: skip ablation TCGA (Runs 25 vs. 08: 95.28% vs. 95.26%); skip ablation PANDA (Runs 26 vs. 10: 37.44% vs. 37.56%). The right-margin verdict column encodes the CI-overlap test, so the status of the skip ablations and the resolvable LR, Macenko, and GradNorm effects are read directly from the plot.

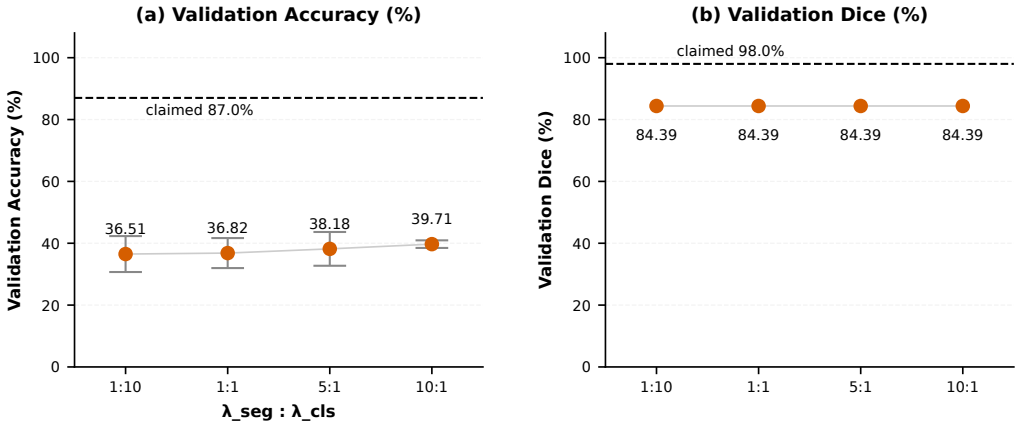


Figure 8: Static loss-weight ratio sweep on PANDA × VGG16 (raw input, $\eta = 10^{-3}$, no GradNorm). (a) Validation accuracy with 95% bootstrap-CI whiskers across the five folds: $\lambda_{seg} : \lambda_{cls}$ = 1:10 (36.51%), 1:1 (36.82%), 5:1 (38.18%), 10:1 (39.71%); the black dashed line marks the claimed 87.0% accuracy. (b) Validation Dice for the same four arms (84.39%, 84.39%, 84.39%, 84.39%) against the claimed 98.0%. The ratios yield comparable accuracy (36.51% – 39.71%), yet no ratio approaches the claimed target — the gap is not a loss-weighting artifact.

yet no ratio approaches the claimed 87% – 88% (Figure 8). The isolated GradNorm arm (Run 18) drops to 28.21% against the 38.18% static baseline with non-overlapping CIs, and the canonical decoupled scheme plateaus at 22.70% (95% bootstrap CI [14.66, 30.73]). Dynamic balancing in either formulation cannot bridge the gap (Figures 6, 7).

5.4. Limitations

Several methodological limitations of this audit should be explicitly considered:

1. *Reconstruction of Underspecified Protocols:* Because the baseline authors released no training code, seed policies, or split manifests, our reproduction reconstructed the pipeline faithfully from published text descriptions and official benchmark repositories. Where protocols were impossible to execute due to missing ground truths (such as the unannotated segmentation masks for 7,421 ISIC 2018 images), we evaluated the established multi-tissue PanNuke benchmark instead.
2. *Thumbnail vs. Whole-Slide Multiscale Analysis:* Our audit directly tested standard 2D multi-task U-Nets on fixed-resolution tiles as spec-

ified in the published recipe. While necessary to evaluate the claims directly, thumbnail-level classification cannot capture the multiscale context required for gigapixel whole-slide grading in computational pathology.

3. *Computational Budget and Seed Scope:* Each of the 27 configurations was evaluated across 5-fold cross-validation under fixed seeds per fold (135 total fold-runs) with an early-stopping patience of 10 epochs. While fold-level bootstrap confidence intervals characterize data-partitioning uncertainty, evaluating multiple random seeds per fold across all 135 models was computationally prohibitive.
4. *Metric Estimator Sensitivity:* As demonstrated in Section 3.2, biomedical segmentation metrics depend heavily on aggregation conventions. We disclosed both batch-pooled validation macro-Dice and per-case empty-credit Dice to make these dependencies transparent.

6. Conclusion & Recommendations

Across 27 configurations $\times$ 5 folds (135 fold-cells), reported near-saturated metrics (88% – 99%) did not reproduce under rigorous cross-validation. The performance gap stems from dataset complexity, task-formulation divergence, and multi-task optimization trade-offs: lateral skip connections provide no resolvable classification benefit, while dynamic gradient balancing ($\alpha = 1.5$) destabilizes classification (38.18% $\rightarrow$ 28.21% Acc). We recommend: (1) mandatory reporting of sample and patient grouping cardinalities, ensuring independent partition units; (2) explicit mathematical definition of the Dice evaluation convention, reporting foreground-only lesion overlap on sparse datasets alongside negative-case false-positive rates; (3) reporting 95% bootstrap confidence intervals on matched ablation comparisons to distinguish architectural effects from fold variance; (4) validating pathology models on multiscale benchmarks rather than low-resolution proxy thumbnails; and (5) systematic evaluation of multi-task balancing hyperparameters before clinical translation.

None of the mechanisms quantified here is exotic. Each is grounded in the evaluation literature (Section 2), and the study whose figures motivated this audit is already being cited as evidence of multi-task efficacy in follow-on oncology work. The recommendations above are offered as a verification checklist for multi-task detection and segmentation models intended for clinical consumption.

Declaration of generative AI and AI-assisted technologies in the writing process

AI-assisted tools were used for language refinement and figure preparation. All scientific content, experiments, data analysis, and conclusions are the sole work of the authors, who reviewed and approved the final manuscript.

CRediT author statement

Muhammad Abdullah Ali: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Project administration. Muhammad Ibrahim Kiani: Writing – review & editing. Muhammad Abdullah Aamir: Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

All training scripts, per-run JSON summaries, evaluation checkpoints, generated figures, and the complete results matrix with 95% bootstrap confidence intervals (`paper_results_matrix_with_ci.csv`) are publicly available at https://github.com/ApatheticMioz/cancer-pathology-dl.

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